

Financial Liberalization and the Effectiveness of Reserve Policy: Evidence from China's 2015 Deposit Rate Reform

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Main Question

Did China's 2015 deposit-rate liberalization strengthen RRR transmission through bank deposit pricing?

Thesis

After the ceiling was removed, deposit rates became more responsive to RRR.

The fall in average deposit rates reflects funding-composition adjustment, not weaker transmission.

U.S. Benchmark: Regulation Q Deregulation

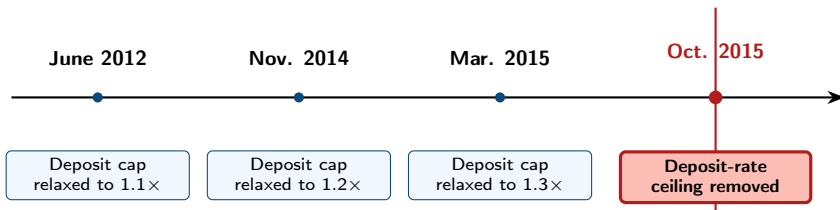
- Regulation Q capped U.S. deposit rates; the 1980 DIDMCA phased out most ceilings by 1986.
- U.S. lesson: removing ceilings intensified deposit competition and raised banks' funding costs.
- The profit-margin channel is central in the U.S. debate: Flannery (1983) asks whether removing ceilings would weaken bank profits; Humphrey and Pulley (1997) find short-run profit-efficiency losses after interest-rate deregulation.
- Contrast: in China, my estimates do not show clear commercial-bank profit-margin compression; funding composition adjusted instead.

References: Gilbert (1986), "Requiem for Regulation Q," *FRB St. Louis Review*; Flannery (1983), "Removing Deposit Rate Ceilings," *FRB Philadelphia Business Review*; Humphrey and Pulley (1997), "Banks' Responses to Deregulation," *JMCB*; Federal Reserve History (2025), "Interest Rate Controls (Regulation Q)."

Why Deposit Ceilings Matter for RRR Transmission

- RRR changes usable deposits.
- Deposit ceilings limited funding-price adjustment.
- Removing the ceiling should restore deposit-rate pass-through.

The 2015 Reform as a Sharp Institutional Cutoff



Earlier reforms relaxed the cap; October 2015 removed it.

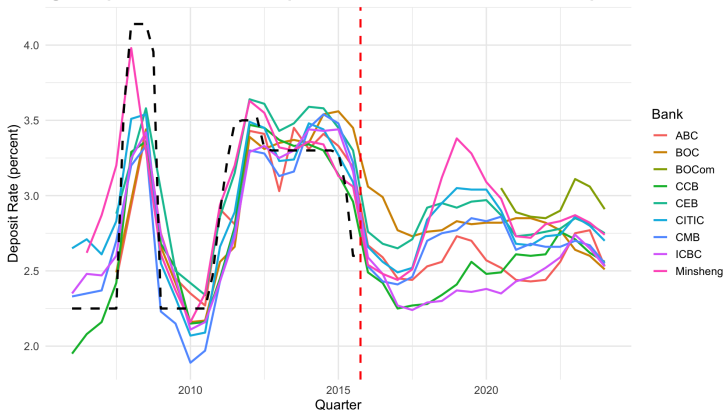
Data

- Quarterly bank-level panel, 2006–2024.
- Sample: major listed Chinese commercial banks.
- Main outcome: effective deposit rate.
- Policy variables: RRR and benchmark rates.

Effective deposit rate = interest expense / total deposits.

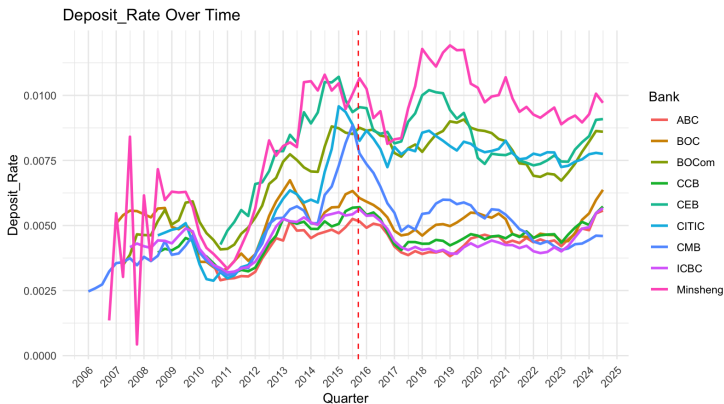
The Ceiling Was Binding Before 2015

Average Corporate Fixed-Term Deposit Rates and Real Benchmark Deposit Rate



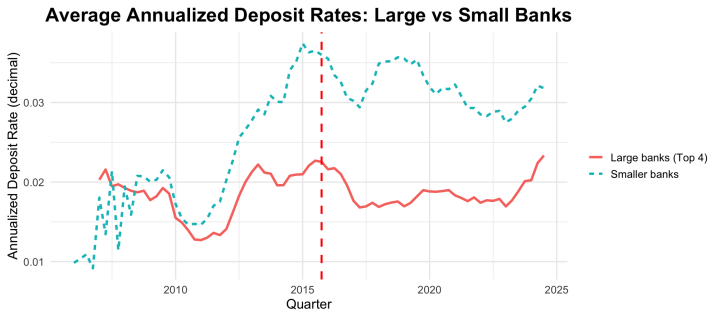
Fixed-term deposit rates clustered near the regulated ceiling before the 2015 removal.

Average Deposit Rate



Average deposit rates fall after liberalization, which motivates the funding-composition mechanism.

Puzzle: Deposit Rates Diverge After Liberalization



After 2015, deposit rates do not simply rise. Large and smaller banks diverge.

Regime-Change Pass-Through Specification

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 RRR_t + \beta_2 Post2015_t + \beta_3 (RRR_t \times Post2015_t) + X'_{it} \delta + \varepsilon_{it}$$

- **Key coefficient:** β_3 .
- β_3 = change in RRR pass-through after ceiling removal.
- Identification idea: a sharp 2015Q4 cutoff creates a discrete change in the RRR-deposit-rate relationship.

Main Result: Full Pass-Through Estimates

Comparison of key coefficients: RRR and RRR $\times$ Post 2015

Top 5–10 Banks					
	Dep. Rate	Loan Rate	ROA	ROE	NIR
Post 2015	−0.2842 (0.2171)	−0.8625 (1.1700)	0.0723 (0.0749)	−0.5957 (0.9482)	−44.16 (33.66)
RRR $\times$ Post 2015	0.0173 (0.0119)	0.0498 (0.0765)	−0.0045 (0.0043)	0.0012 (0.0526)	2.464 (1.865)
Top 4 Banks					
Post 2015	−0.8466 (0.0834)**	0.1578 (0.1760)	0.0584 (0.0271)	0.2886 (0.9413)	32.51 (3.671)**
RRR $\times$ Post 2015	0.0487 (0.0047)**	0.0067 (0.0136)	−0.0054 (0.0019)	−0.0853 (0.0521)	−1.663 (0.2356)**

Standard errors in parentheses. ** $p < 0.01$, * $p < 0.1$.

Interpretation: Pass-Through Restored, Rates Fell

- Core result: liberalization changed how deposit rates respond to RRR.
- Negative level effect → puzzle.
- Positive interaction → restored pass-through.

Validation: The Break Is Concentrated Around 2015Q4

- Event study: pass-through shifts around the reform date.
- Placebo cutoffs: fake pre-2015 dates do not explain the core result.
- Exclusion windows: results survive dropping anticipation periods.
- MLF controls and bank trends: main result remains stable.

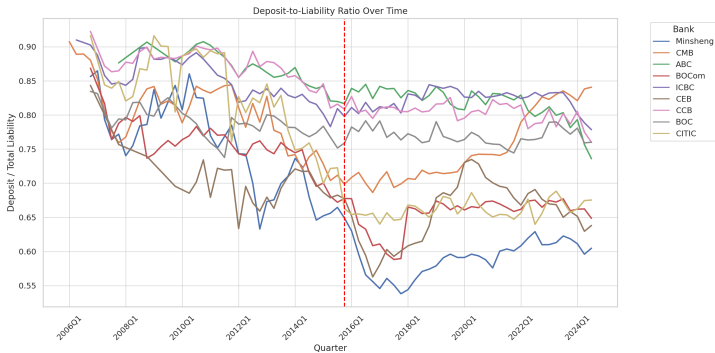
Why Did Average Deposit Rates Fall?

If pass-through was restored, why did average deposit rates fall?

Answer

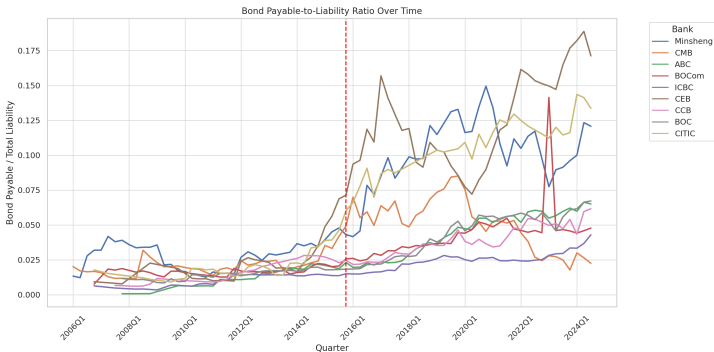
Liberalization changed banks' funding choices, not only their posted deposit rates.

Mechanism I: Banks Reduced Reliance on Deposits



After 2015, several banks reduced deposits as a share of liabilities.

Mechanism II: Bond and Wholesale Funding Became More Important



Bond and wholesale funding became more important after the reform.

Price Adjustment vs Funding-Composition Adjustment

- Large banks: stronger direct deposit-rate adjustment.
- Smaller banks: stronger funding-composition adjustment.
- Deposit rates can fall when alternative funding lowers the marginal value of deposits.

What the Model Needs to Explain

- 1 Binding ceiling mutes RRR pass-through.
- 2 Removing the ceiling restores pass-through.
- 3 Alternative funding can lower average deposit rates.
- 4 Bank heterogeneity shapes adjustment.

Baseline Model: A Binding Ceiling Mutes RRR Pass-Through

$$\max_{r \leq \bar{r}} \pi(r) = f((1 - \alpha)a(r)) - ra(r)$$

$$\text{If } r^* = \bar{r}, \quad \frac{dr^*}{d\alpha} = 0.$$

Implication

RRR can change shadow values, but observed deposit rates cannot adjust.

After Liberalization: RRR Can Move Deposit Prices

Let

$$\Phi(r, \alpha) = f'((1 - \alpha)a)(1 - \alpha)a_r - [a + ra_r], \quad x = (1 - \alpha)a(r).$$

$$\frac{dr^*}{d\alpha} = -\frac{\partial\Phi/\partial\alpha}{\partial\Phi/\partial r}, \quad \frac{\partial\Phi}{\partial r} < 0.$$

$$\frac{\partial\Phi}{\partial\alpha} = -a_r [f'(x) + xf''(x)].$$

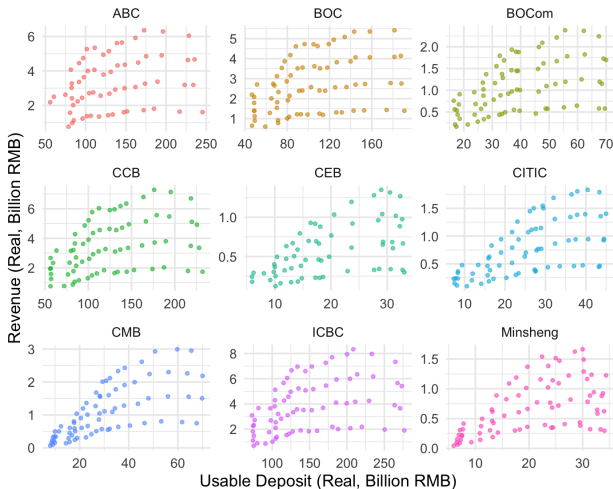
$$\frac{dr^*}{d\alpha} > 0 \iff f'(x) + xf''(x) < 0 \iff \frac{f'(x)}{xf''(x)} > -1 \quad (f''(x) < 0).$$

Empirical counterpart

Positive RRR $\times$ Post 2015 coefficient.

Curvature Check

CPI-Normalized Revenue vs. Usable Deposit (in Billions)



The revenue function is concave enough for the positive RRR comparative static.

Calibration: Latent Optimal Rates Under the Dynamic Model

Actual vs. Counterfactual Deposit Rates by Bank ($\beta = 0.99$)



$\beta = 0.99$

Actual vs. Counterfactual Deposit Rates by Bank ($\beta = 0.95$)



$\beta = 0.95$

Counterfactual rates are structural benchmarks from the dynamic model, not forecasts.

Extension: Flexible- and Fixed-Term Deposits and Bonds

Banks can raise flexible-term (S) and fixed-term (L) deposits with different retention $\delta_S < \delta_L$, and issue wholesale bonds B .

Steady-state structure:

$$D_j^* = \frac{a_j(r_j, i_B)}{1 - \delta_j}, \quad f'(X^*) = i_B, \quad X^* = (1 - \alpha)(D_S^* + D_L^*) + B^*.$$

FOC for each deposit type $j \in \{S, L\}$:

$$a_{jr}(r_j, i_B)(1 - \alpha) \frac{f'(X^*)}{1 - \beta\delta_j} = a_j(r_j, i_B) + r_j a_{jr}(r_j, i_B).$$

Comparative statics:

$$\frac{\partial D_S^*}{\partial \delta_S} > 0, \quad \frac{\partial B^*}{\partial \delta_S} < 0.$$

- Less stable deposits imply more bond and longer-term funding.
- Liberalization reveals heterogeneous funding strategies.

One-Slide Summary

- Before 2015: deposit-rate ceiling muted RRR pass-through.
- After 2015: deposit rates became more responsive to RRR.
- Puzzle: average deposit rates fell.
- Mechanism: banks adjusted liability composition.
- Takeaway: transmission depends on bank funding structure.

Conclusion

- 1 Liberalization restored the RRR price channel.
- 2 Deposit rates became more responsive to RRR after 2015.
- 3 The fall in average deposit rates reflects funding substitution, not failure of transmission.
- 4 Bank liability structure determines how strongly RRR transmits after liberalization.

Final message

Financial liberalization made reserve policy more powerful through market pricing, but also made transmission depend more on banks' funding structure.

Thank you

Questions and comments are welcome.

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Backup: Broader PBOC Toolkit

- Reserve Requirement Ratio (RRR) is the liquidity quantity control.
- Benchmark deposit and loan rates are administrative pricing anchors.
- Open Market Operations handle short-term liquidity operations.
- Medium-Term Lending Facility supplies medium-term liquidity after 2015.
- Loan Prime Rate is the post-2019 lending benchmark linked to MLF.

Backup: Literature Position

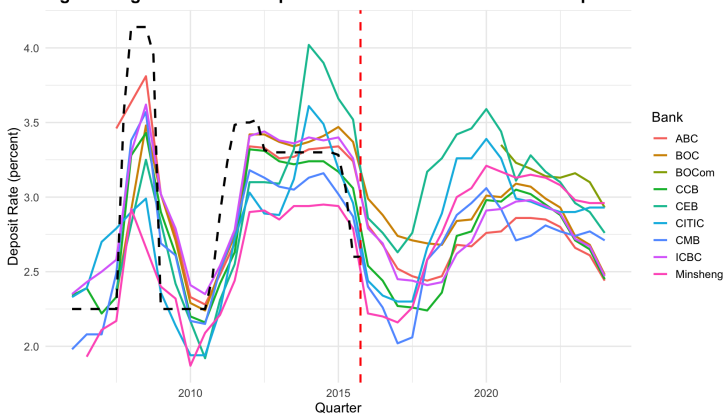
- Existing work studies interest-rate liberalization and monetary transmission.
- Research on China emphasizes RRR as a liquidity and credit-management tool.
- International evidence on deposit-rate ceilings shows funding-cost effects.
- This paper focuses on RRR pass-through through bank funding costs.

Backup: Identification and Scope

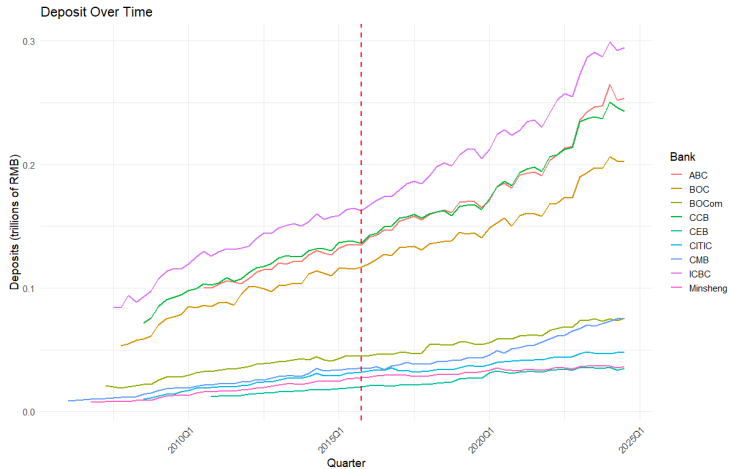
- The reform applies system-wide, so the design is not a treated-control DID.
- I interpret the specification as a regime-change pass-through design.
- Key idea: absent the reform, the RRR-deposit-rate relationship would not break discretely at 2015Q4.
- Support: event-study timing, placebo cutoffs, exclusion windows, bank trends, and MLF controls.
- Interpretation is strongest for deposit-rate pass-through.

Backup: Fixed-Term Saving Deposit Rate

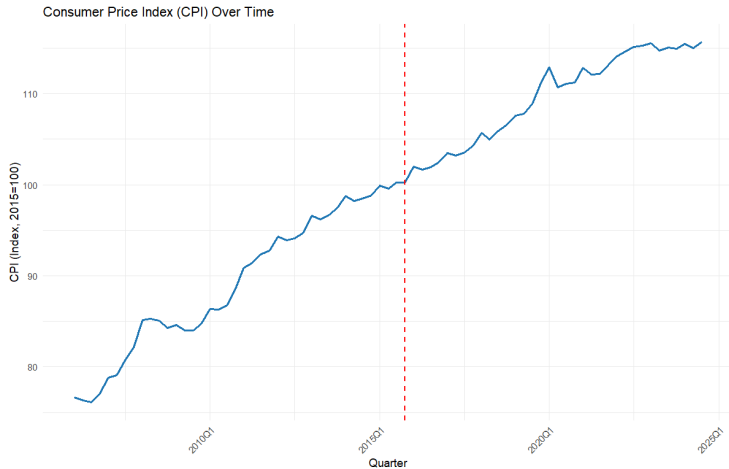
Average savings Fixed-Term Deposit Rates and Real Benchmark Deposit Rate



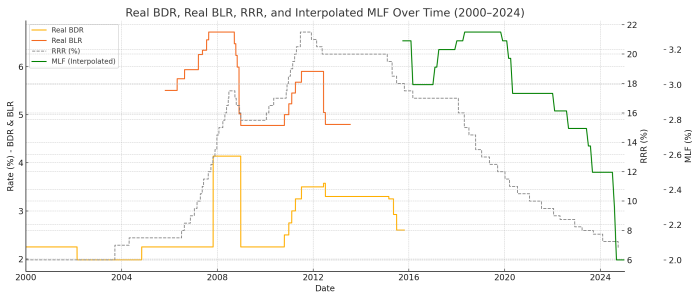
Backup: Total Deposits



Backup: CPI

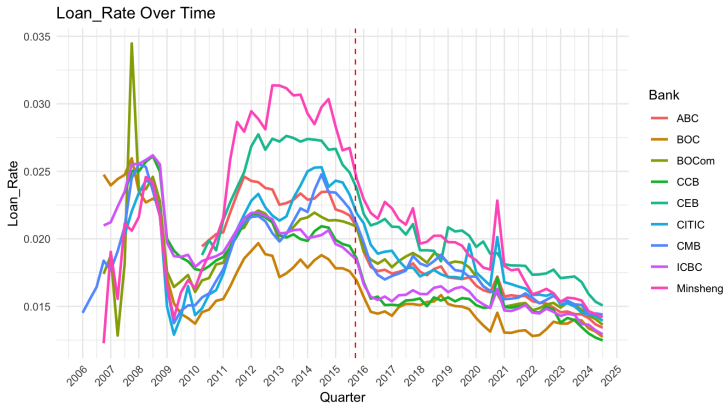


Backup: RRR, Benchmark Rates, and MLF

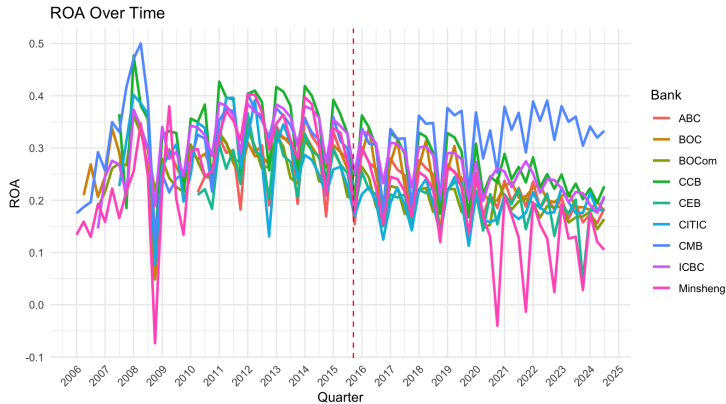


Source: PBOC.

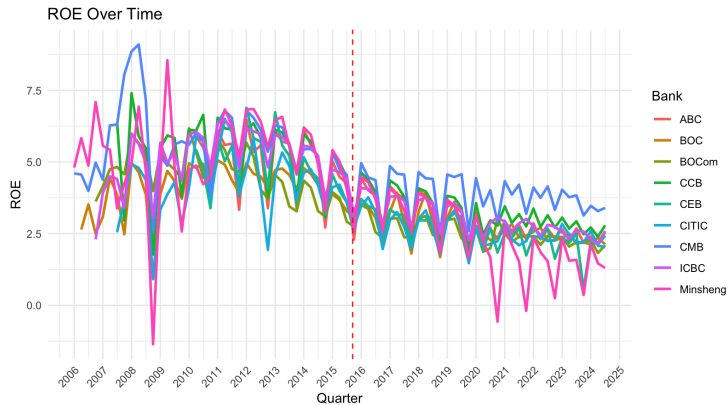
Backup: Loan Rate



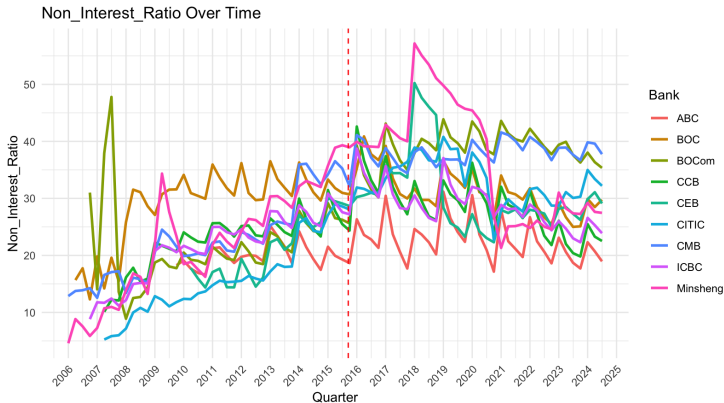
Backup: ROA



Backup: ROE



Backup: Non-Interest Income Ratio



Backup: Dynamic Deposit Accumulation

Deposit evolution:

$$D_{t+1} = \delta D_t + a(r_t), \quad 0 \leq \delta < 1.$$

Bank objective:

$$\max_{\{r_t\}} \sum_{t=0}^{\infty} \beta^t [f((1-\alpha)(\delta D_t + a(r_t))) - r_t a(r_t)].$$

Steady state and FOC:

$$D^* = \frac{a(r^*)}{1-\delta}, \quad f'((1-\alpha)D^*)(1-\alpha) \frac{1}{1-\beta\delta} a_r = a + r a_r.$$

- Deposit stickiness dampens immediate RRR effects.
- Once the ceiling is removed, higher $dr^*/d\alpha$ implies stronger policy transmission.

Backup: Revenue Curvature Test

Estimated revenue function:

$$\text{Revenue}_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 x_{it}^2 + \varepsilon_{it}, \quad x = (1 - \alpha) \cdot \text{Deposit}.$$

Curvature condition:

$$\frac{f'(x)}{|f''(x)|} > x \iff \frac{\beta_1 + 2\beta_2 x}{2|\beta_2|} > x.$$

- Use delta method to compute the standard error of the test statistic.
- Evaluate at the sample mean x_0 for each bank.
- Details support the sufficient curvature condition used in the model.

Backup: Curvature Test Results

Bank	Curvature Ratio	Std. Error	Z-Score	p-value	Strong Curvature
CMBC	-5.99×10^9	2.02×10^9	-4961	$< 10^{-10}$	Yes
CMB	-3.06×10^{10}	8.24×10^9	-1217	$< 10^{-10}$	Yes
ABC	-5.36×10^{10}	2.68×10^{10}	-375	$< 10^{-10}$	Yes
BOCOM	-1.91×10^{10}	7.38×10^9	-1358	$< 10^{-10}$	Yes
ICBC	-5.92×10^{10}	1.66×10^{10}	-606	$< 10^{-10}$	Yes
CEB	-1.20×10^{10}	7.11×10^9	-1408	$< 10^{-10}$	Yes
CCB	-5.27×10^{10}	1.75×10^{10}	-576	$< 10^{-10}$	Yes
BOC	-4.85×10^{10}	2.15×10^{10}	-468	$< 10^{-10}$	Yes
CITIC	-2.25×10^{10}	1.34×10^{10}	-746	$< 10^{-10}$	Yes